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Voices of the crowd: Exploring user evaluations of ChatGPT using structural topic modeling

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ABSTRACT

ChatGPT represents a significant advancement in Human–Computer Interaction, yet public perceptions of the technology remain divided and insufficiently documented. This study analyzes large-scale YouTube comments posted between 2023 and 2025 to capture how general users evaluate ChatGPT's capabilities, limitations, and societal implications. Using a computational mixed-methods approach that integrates sentiment analysis with Structural Topic Modeling, this study reveals a clear contrast between technical definitions of system behavior and how users interpret it. Perceived bias is conceptualized less in terms of statistical characteristics and more as a matter of political or ideological significance, aligning with common folk theories about algorithmic behavior. The findings also identify a utility–anxiety tension: while users praise ChatGPT for improving efficiency in programming, structured reasoning, and creative writing, they simultaneously express concerns about factual reliability, content authenticity, and the potential displacement of both low-skill and knowledge-based jobs. Importantly, negative evaluations stem from specific encounters with error or inconsistency rather than generalized resistance to AI, whereas positive evaluations center on clear instrumental gains. The observed patterns indicate that attitudes among users are determined by particular circumstances and direct engagements, rather than by abstract assumptions. Overall, the study provides a psychologically informed account of public interpretation of generative AI, offering insights that can guide developers and policymakers in aligning system design, governance practices, and risk communication with the lived experiences and concerns of general users.

1. Introduction

Artificial intelligence (AI) emerged as a formal discipline at the 1956 Dartmouth Conference and has since progressed through successive waves of theoretical and technological innovation. The resurgence of AI around 2010, driven primarily by advances in deep learning, enabled rapid expansion in natural language processing (NLP), speech recognition, and pattern recognition (Haenlein & Kaplan, 2019). Building on this foundational work, OpenAI launched ChatGPT in November 2022, which represents a substantive leap forward in both the design paradigm and functional capabilities of chatbot technology (Skjuve et al., 2021). With its high level of language proficiency and fluid, human-like conversational interface, ChatGPT has significantly reshaped human–computer interaction (HCI) and broadened the range of tasks that can be mediated through AI (Islam et al., 2025; Pham et al., 2025; Stokel-Walker & Van Noorden, 2023; Tekinay, 2023). As the system continues to integrate into professional, educational, and everyday environments, examining its evolving role offers a timely perspective on how contemporary AI technologies are transforming communication and interaction practices (Baker & Utku, 2025).

The current body of ChatGPT research primarily centers around its application scenarios, technological advancements, and societal implications from an institutional or expert perspective. Scholars have highlighted how ChatGPT's dialogue capabilities and content generation potential open up new opportunities across various domains, such as digital government (Papagiannidis et al., 2025), educational ecology (Fergus et al., 2023; Mogali, 2023), technological innovation (Homolak, 2023), and artistic creation (McGee, 2023). For instance, Gordijn and

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Have (2023) suggest that ChatGPT could enhance researchers' efficiency. Moreover, a meta-analysis involving 2908 participants revealed that AI significantly influences academic achievement (Zheng et al., 2021). Additionally, studies indicate that AI exerts a substantial influence on higher education outcomes (Wu & Yu, 2023). Although these studies have shed light on the potential value of ChatGPT in professional settings, they often fail to account for the detailed, subtle interactions involving non-expert users—interactions that are a key feature of ChatGPT's widespread use. Furthermore, studies identifying risks such as over-reliance on automated outputs and security vulnerabilities in medical practice (Corsello & Santangelo, 2023) tend to reflect top-down concerns centered on professional responsibilities, rather than the everyday anxieties expressed by ordinary users.

From a technical perspective, research has focused on model architecture and training. ChatGPT builds upon the Transformer model, adopting a pre-training approach using extensive corpora to enhance natural language understanding (Bhattacharya et al., 2023; Eysenbach, 2023). While technical investigations into dialogue quality and flow control aim to improve conversational effectiveness (Cascella et al., 2023), there is a disconnect between these metric-based technical evaluations and the subjective, lived experience of the end-user. Driven by these advancements, ChatGPT has yielded profound social impacts in the AI industry (Kung et al., 2023) and labor employment (Santo & Joviano-Santos, 2023). However, a comprehensive understanding of ChatGPT's implications remains incomplete without analyzing how these technical complexities are perceived by the lay public.

Despite the changes that ChatGPT has brought to people's lives, there are significant differences in users' perceptions of its use. For instance, a study on medical students showed that they have a positive basic attitude toward AI and medical chatbots, but the students also expressed concerns (Moldt et al., 2023). Moreover, research has found that users with a high social orientation have a more positive attitude toward chatbots (Maar et al., 2023). Another study focused on teachers' attitudes toward educational chatbots, suggesting that perceived ease of use and usefulness lead to greater acceptance of chatbots, and that the formal language of chatbots leads to higher intent to use them (Chocarro et al., 2023). Although Zhu (2025) examined users' perceptions of ChatGPT, the study primarily focuses on technically oriented learners engaged in programming tasks, resulting in a limited scope that does not capture the broader range of public evaluations. It is clear that public perception of AI is complex and diverse, but there is little research on public perception and attitudes toward ChatGPT. Additionally, previous research mostly used case study methods to analyze the perception of a specific group, but due to the singularity of the survey subjects, there is no clear understanding of the distribution of public perceptions and attitudes toward ChatGPT. Therefore, this study aims to explore the key attributes of ChatGPT that users value, while also identifying the factors contributing to both their satisfaction and concerns regarding its usage.

To address the limitations of prior research that relies on expert commentary or narrowly defined user groups, this study draws on large-scale UGC to develop a systematic account of everyday user perceptions. Unlike the restricted datasets analyzed by Jeong and Sung (2025) or Xu et al. (2024), our longitudinal corpus of YouTube comments captures diverse public discussions over an extended period. This approach allows the findings to reflect routine, non-expert experiences rather than the more insulated perspectives of professional or institutional actors. Methodologically, and differing from Jeong and Sung (2025), the use of Structural Topic Modeling (STM) with sentiment incorporated as a covariate extends existing analytical frameworks by enabling thematic patterns and evaluative stance to be examined simultaneously. This design reveals that negative sentiment tends to cluster around systemic risks, while positive sentiment is closely associated with the technology's practical utility. The topic correlation network further clarifies how user discussions organize into coherent structures. Collectively, this study contributes by offering a more detailed and comprehensive understanding of the social reception of ChatGPT.

2. Literature review

2.1. Introduction to ChatGPT

In the modern era of AI, the rapid advancement of NLP technology offers us new opportunities for understanding and generating language. NLP is a subfield of computer science, with its origins dating back to the 1950s, that has gone through several stages, including rule-based, statistics-based, and deep learning-based, exploring how computers can understand and generate natural language. It is a vital component of AI and plays a critical role in the development of chatbots. Particularly, ChatGPT from OpenAI, a member of the Generative Pretrained Transformer (GPT) series of models, has received substantial attention for its optimized performance in conversational scenarios, providing a natural interaction experience for users.

The superior performance of ChatGPT has enriched users' perceptions of AI. The perception of AI among users is a complex and multidimensional concept that scholars have explored from multiple perspectives. Firstly, some researchers have defined users' perceptions of AI as subjective experiences and cognitions of AI's technical characteristics, functionalities, effects, and impacts, including perceptions in various situations and evaluations of user experience (Luger & Sellen, 2016). However, Assuncao et al. (2022) pointed out that this definition overlooks users' emotional experiences and reactions; therefore, they propose to include emotion as part of users' perceptions of AI, thus broadening the understanding of this concept.

Secondly, users' perceptions of AI are based on an interactive socialization process. This process involves the interaction between users and AI technology, as well as the influence of application scenarios, thereby shaping users' attitudes, beliefs, emotions, and behavioral intentions toward AI (Lee & LaRose, 2011). Building on this, some researchers believe that users' perceptions of AI should be defined as the cognitive, emotional, and behavioral reactions generated by users when interacting with it in application scenarios. They further propose that users' perceptions of AI can be regarded as social identities, social influences, and social relations formed during interactions with AI in application scenarios, which are influenced by AI's characteristics. Therefore, this study believes that users' perceptions of AI are formed in the process of interacting with AI technology, including a comprehensive experience of cognitions of AI characteristics, emotional reactions, perceptions of the interaction environment, and behavioral responses (Assuncao et al., 2022; Van der Velden & Mörtberg, 2015).

2.2. Public perception of AI

Research on public perceptions of AI has long emphasized how individuals understand, evaluate, and emotionally respond to emerging technologies. Early studies documented generally optimistic attitudes, with many people expecting AI to enhance efficiency and convenience in daily life (McClure, 2017). Yet this optimism has always coexisted with apprehension. Concerns about job displacement, loss of human control, and long-term societal risks have been persistent themes, particularly as AI systems become more capable and embedded in workplace routines (Zhang & Dafoe, 2019).

As public awareness of AI deepened, attention gradually shifted toward ethical and governance considerations. Scholars found that individuals increasingly worry about privacy breaches, data misuse, and security threats, reflecting heightened sensitivity to the risks associated with algorithmic decision-making (Jazim et al., 2025). Cross-national comparisons offer additional details about how these concerns vary across cultural contexts. For example, Zhang and Dafoe (2019) report that the American public expresses notable caution regarding AI's implications for employment and personal privacy, while Chinese respondents tend to exhibit more optimism. Similarly, Cui and Wu (2019) observe that Chinese participants largely believe humans will retain control over AI, though they also acknowledge risks such as data leakage, technological unemployment, and weakened interpersonal relationships.

Building on these broader patterns of AI perception, recent studies have examined public attitudes specifically toward Generative AI (GenAI) systems such as ChatGPT. This research consistently finds that sentiment on social platforms remains predominantly positive but is shaped by platform characteristics, regional contexts, and shifting public attention over time. Demirel et al. (2024), using geolocation-integrated semantic network analysis of Twitter, identify clear geopolitical differences: users in the Global North emphasize technical specifications and sector-specific integration, whereas those in the Global South focus on cultural and linguistic adaptation and privacy concerns. Jeong and Sung (2025), analyzing South Korean YouTube news comments, find that public discourse centers on labor market transformations, fears of losing control over AI, and increasing interest in human-centered and educational applications. Xu et al. (2024), studying Reddit discussions, report that 61.6% of comments express positive sentiment, organized around themes of user experience, technical capabilities, and broader societal impact, with sentiment fluctuations tied to major product releases such as GPT-4 and Code Interpreter.

2.3. Perception of AI tools by users

Researchers have examined users' perceptions of chatbots across a wide range of dimensions, including application scenarios, system characteristics, user attributes, emotional responses, and interaction contexts (Demirel et al., 2024; Eysenbach, 2023; Zhang et al., 2025). These perceptions play a central role in shaping users' emotional reactions, cognitive evaluations, and subsequent behavioral intentions (Bickmore & Cassell, 2001). In task-oriented settings such as customer service or education, users tend to view chatbots as collaborative partners for problem-solving or knowledge acquisition (Følstad et al., 2021). In these contexts, effective systems typically demonstrate competence, comprehension, and an approachable communication style to support constructive engagement (Winkler & Soellner, 2018). By contrast, in entertainment and gaming environments, users often approach chatbots as competitive agents rather than cooperative counterparts (Tsay-Vogel et al., 2018). Here, attributes such as skillfulness, fairness, and behavioral predictability are essential for sustaining user interest and delivering an engaging interactive experience (Yee et al., 2009).

Users' cognition and emotions toward chatbots also influence their behavioral tendencies. Users who perceive a chatbot as highly intelligent and reliable are likelier to cooperate with it (Luger & Sellen, 2016). Conversely, negative emotions toward a chatbot may lead to resistance and impact user behavior (Brandtzaeg & Følstad, 2017; Zhang et al., 2022). Furthermore, social norms and cultural factors play a significant role in user interactions with chatbot AIs. Politeness may be emphasized in conversations with chatbots within certain cultural contexts (Lim, Rooksby, & Cross, 2020), suggesting that users' perceptions and expectations of chatbot AI vary across cultures, influencing their behaviors and interaction styles. Privacy is another critical factor shaping users' perceptions of chatbot AI. Users generally trust chatbots that prioritize privacy protection (Nordberg et al., 2020). Privacy concerns affect not only the trustworthiness of chatbot AI but also individuals' trust in others within the social environment. Hence, the inherent features of AI technology significantly influence its acceptance (Chen et al., 2024; Ma et al., 2025).

Regarding chatbot characteristics, system features like language style have been investigated for their impact on user perceptions (Davis, 1993). For instance, AI systems displaying politeness and casual language are perceived as more empathetic. However, the effects may vary depending on the application context and user preferences. While advancements in NLP enable more flexible and contextual interactions, unclear system limitations can lead to "conversational deception" or misunderstandings. The personality of the chatbot also influences users' perceptions. Research indicates that chatbots with affable and humorous personalities receive higher ratings from users (Bickmore & Cassell, 2001). Moreover, users' own characteristics, including gender, age, and cultural background, impact their perceptions of chatbots. Female users, for example, tend to prefer interacting with cuter chatbots (Zhang et al., 2022), while younger users favor chatbots with distinct personalities (Bickmore & Cassell, 2001). Cultural backgrounds also shape users' expectations and preferences for chatbots (Lee et al., 2010).

3. Methodology

3.1. Research design

This study adopts a pragmatist research paradigm, which emphasizes selecting methods that best address the research question rather than adhering to a single methodological tradition (Creswell & Plano Clark, 2018). Because the dataset consists of large volumes of unstructured text, an exclusively qualitative approach would be difficult to implement, while a strictly quantitative approach would overlook important differences in how users express their views. To balance these strengths and limitations, the study applies a computational mixed-methods design that combines large-scale text analysis with interpretive examination of key themes (Lewis, 2009). The overall research workflow is presented in Fig. 1.

3.2. Data collection

The dataset for this study was derived from YouTube, one of the world's most widely used platforms for user-generated content (UGC) and public discourse. To ensure that the collected material was directly relevant to the research focus, the keyword "ChatGPT" was entered into YouTube's search function. To enhance the relevance and representativeness of the samples, we have also established additional screening criteria. Only those videos with high audience engagement, as measured by view counts, likes, and comment activities, will be included. All user comments were retrieved through a custom-built Python web crawler. The crawler was programmed to collect publicly available comments while excluding metadata that could contain personally identifiable information. To avoid duplication and ensure dataset integrity, comments were cleaned and standardized during extraction. The final dataset includes 201,887 comments from 1130 videos, spanning two years from January 2023 to January 2025. This time frame was selected to cover the launch of ChatGPT, its rapid popularization, and its early integration into public and professional spheres. YouTube channels. YouTube has become a highly popular platform for individuals to share their opinions on a wide range of topics, making it a valuable source of UGC for academic research. Studies across various disciplines have increasingly relied on YouTube's rich repository of UGC to gain insights into public opinion and behavior (Hassani et al., 2020; Ko, 2020; Lim, Kim, et al., 2020).

3.3. Data analysis

The data analysis proceeded in three stages: text preparation, topic estimation, and interpretation. In the first stage, the comments were preprocessed to remove noise and standardize the text for computational analysis. We then identified the optimal number of topics by evaluating a range of candidate models and selecting the specification that best captured the recurring patterns in the corpus. Finally, we examined the resulting topics, assessed their substantive coherence, and interpreted the broader patterns reflected in user discussions.

In selecting a topic modeling approach, we considered the specific challenges posed by social media data, particularly short-text sparsity, where limited word counts reduce the likelihood of meaningful term co-occurrences (Hong & Davison, 2010). Although embedding-based methods such as BERTopic are well-suited to capturing semantic similarity in brief comments, our research objective required a model

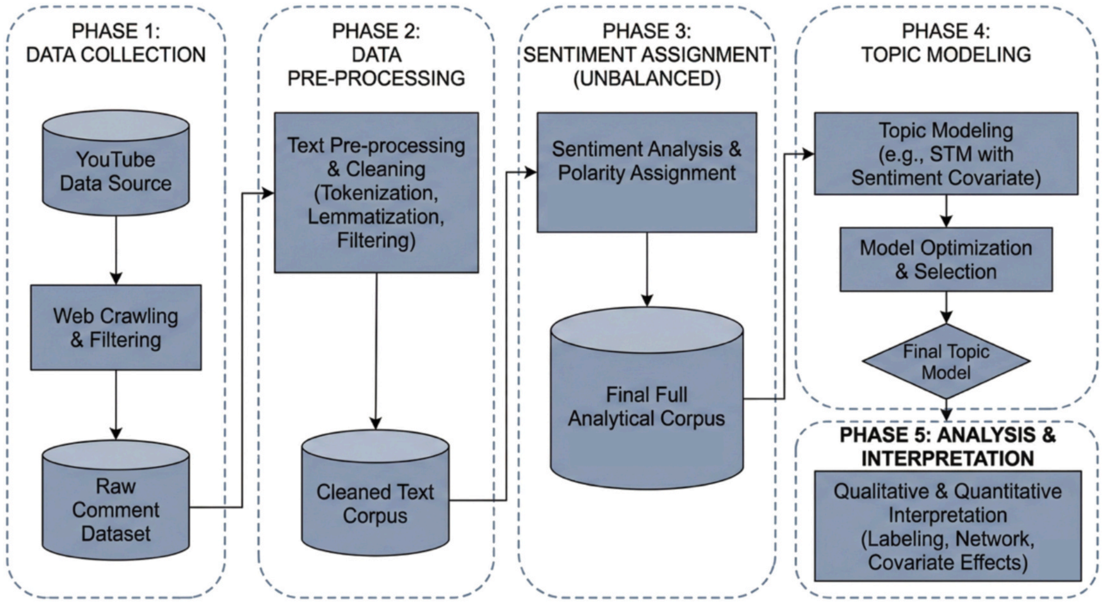

Fig. 1. Research process map.

capable of incorporating document-level covariates. STM was therefore selected because it allows sentiment polarity to be integrated directly into the topic estimation process (Roberts et al., 2016), which allows us to examine how evaluative stance influences the prevalence of specific topics rather than simply clustering comments based on semantic proximity. To manage sparsity without altering the underlying structure of the dataset, comments were retained at the individual level. Instead of aggregating texts, sparsity was addressed through preprocessing, including the removal of short tokens and very low-frequency terms, resulting in a more compact and analytically robust vocabulary suitable for topic estimation.

3.3.1. Preprocessing of text

To ensure analytical rigor, we implemented a systematic preprocessing pipeline designed to minimize noise and standardize textual content within the raw dataset (Hassani et al., 2020). The process began by filtering out non-English comments to establish linguistic consistency. The remaining text was then tokenized into individual semantic units, enabling word-level computation. To further isolate meaningful linguistic content, numeric values, punctuation, and special characters were removed. Vocabulary was subsequently refined through stop-word elimination using the SMART list (e.g., "the," "and") and by discarding tokens with fewer than three characters to reduce sparsity. Then we applied lemmatization to consolidate inflected forms into their root representations, thereby enhancing comparability across terms. We also excluded extremely low-frequency words (occurring in less than 1% of the corpus) and reviews containing fewer than eight terms. While these inclusion criteria resulted in a substantial reduction from the initial raw dataset, this filtering was critical to eliminate low-quality, short, or non-substantive entries.

Following preprocessing, sentiment analysis was conducted using the Bing lexicon. The final analytic corpus comprises 81,103 comments, distributed as 56,950 positive, 12,870 negative, and 11,283 neutral entries. While neutral comments were retained during the topic estimation stage to preserve the semantic structure of technical and factual discussions, the subsequent topic distribution analysis focuses exclusively on the binary opposition between positive and negative polarity. This methodological choice is grounded in the necessity of isolating explicit evaluative stance from descriptive noise; neutral comments in UGC often consist of factual inquiries, short phatic expressions, or ambiguous statements that lack clear attitudinal direction (Liu, 2022).

3.3.2. Estimating the number of topics

The topic modeling analysis is performed using the STM package in the R programming language. The fundamental approach of the STM involves assigning document vectors, each representing a comment (denoted as r), to K topics, with the document vectors comprising a vocabulary of size V. As outlined by Roberts et al. (2019), the selection of the appropriate number of topics is guided by three criteria: (1) Held-out likelihood, which quantifies how well a given number of topics can explain the overall variability in the corpus; (2) semantic consistency of words within each topic; and (3) exclusivity of topic words within each topic (Roberts et al., 2016).

To determine the preliminary number of topics, we began by estimating an initial set of eight topics. We then assessed the held-out likelihood by incrementally increasing the number of topics by two at each step, exploring a range up to a maximum of 40 topics. Finally, we evaluated the topics using the FREX (Frequency-Exclusivity) metric to identify the number that yielded the highest held-out likelihood. The FREX metric is modeled as follows:

$$\mathrm{FREX}_{k,v} = \left(\frac{\omega}{\mathrm{F}\left(\beta_{k,v} \Big/ \sum_{j=1}^{K} \beta_{j,v} \right)} + \frac{1-\omega}{\mathrm{F}\left(\beta_{k,v}\right)} \right)^{-1} \tag{1}$$

where k is the k-th topic, v is the word under consideration, β is the word distribution of the k-th topic, and ω is used to enforce exclusivity (defaulted to 0.7). FREX is estimated as a weighted harmonic average of a word's exclusivity and semantic coherence. Semantic coherence, proposed by Mimno et al. (2011), uses the co-occurrence frequency of the most likely words in each topic, while exclusivity considers the mutual appearance of the most likely words in multiple topics.

As shown in Fig. 2, model fit improved rapidly up to approximately 20 topics, after which gains became marginal, and residual structure flattened, indicating that additional topics no longer captured meaningful semantic variance. To assess topic-level quality, we further compared coherence and exclusivity for models with K = 15–20. Based on Fig. 3, the coherence–exclusivity trade-off showed that the 20-topic solution produced the largest concentration of topics exhibiting both high internal coherence and strong exclusivity, whereas smaller models exhibited theme compression and intermediate specifications displayed greater instability. Taken together, these diagnostics converge in

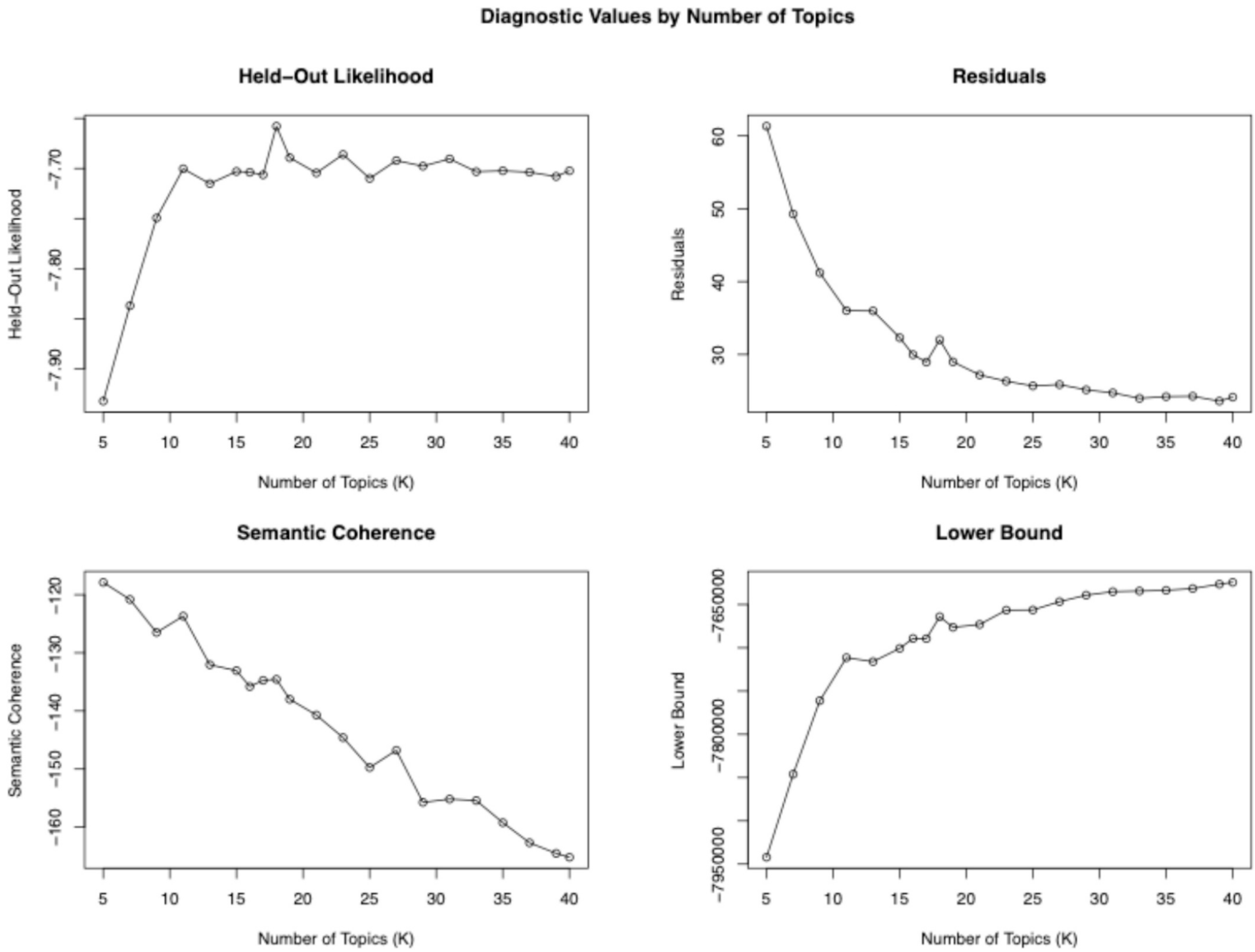

Fig. 2. Metrics of different topic model solutions.

indicating that K = 20 offers the most balanced specification, yielding interpretable and distinct topics without overfitting.

4. Research finding

4.1. Topic labeling

To assign semantic meaning to the algorithmic clusters, this study employed an inductive thematic coding approach (Braun & Clarke, 2006), allowing themes to emerge directly from the data rather than relying on pre-existing theories. The coding process utilized a triangulated review of data for each topic, consisting of the top 10 representative comments, the highest probability words, and the FREX metric. While probability scores identified dominant terms, the FREX metric was explicitly used to isolate distinctive terms that differentiate a specific topic from the broader corpus (Roberts et al., 2016). To ensure methodological rigor and mitigate subjective bias, we applied investigator triangulation; each researcher independently conducted open coding to propose initial labels based on the contextual nuances of the representative comments. These independent codes were subsequently synthesized through an iterative consensus-building session to resolve discrepancies, resulting in the final topic taxonomy presented in Table 1.

4.2. Topic interpretation

Topic 1 captures user evaluation of the model in rule-based game environments. The high-probability words “game,” “play,” “move,” “rule,” and “chess” suggest that the topic is organized around structured gameplay. The FREX words “chess,” “stockfish,” “queen,” and “pawn” further indicate that chess is the primary domain through which users assess the model's game logic and decision quality.

Topic 2 centers on software development activities. High-probability terms such as “code,” “program,” and “error” clearly situate the discourse within programming contexts, while FREX terms like “python,” “fix,” and “overflow” point to specific tasks involving debugging, syntax correction, and language-specific assistance. The frequent occurrence of “error” and “fix” signals persistent challenges in automated code generation and highlights the continued importance of a Human-in-the-Loop workflow. From the representative reviews, we found that users recognize that while AI can accelerate routine development tasks, human oversight remains essential for identifying hallucinated logic, ensuring correctness, and maintaining software reliability.

Topic 3 articulates socio-economic anxiety regarding the automation of cognitive labor. High-probability terms like “job” and “replace” signal general concern, while FREX terms such as “engineer,” “designer,” and “automation” specifically identify knowledge workers as vulnerable to automation. Several studies have shown that AI could automate many tasks currently performed by human workers, leading to job

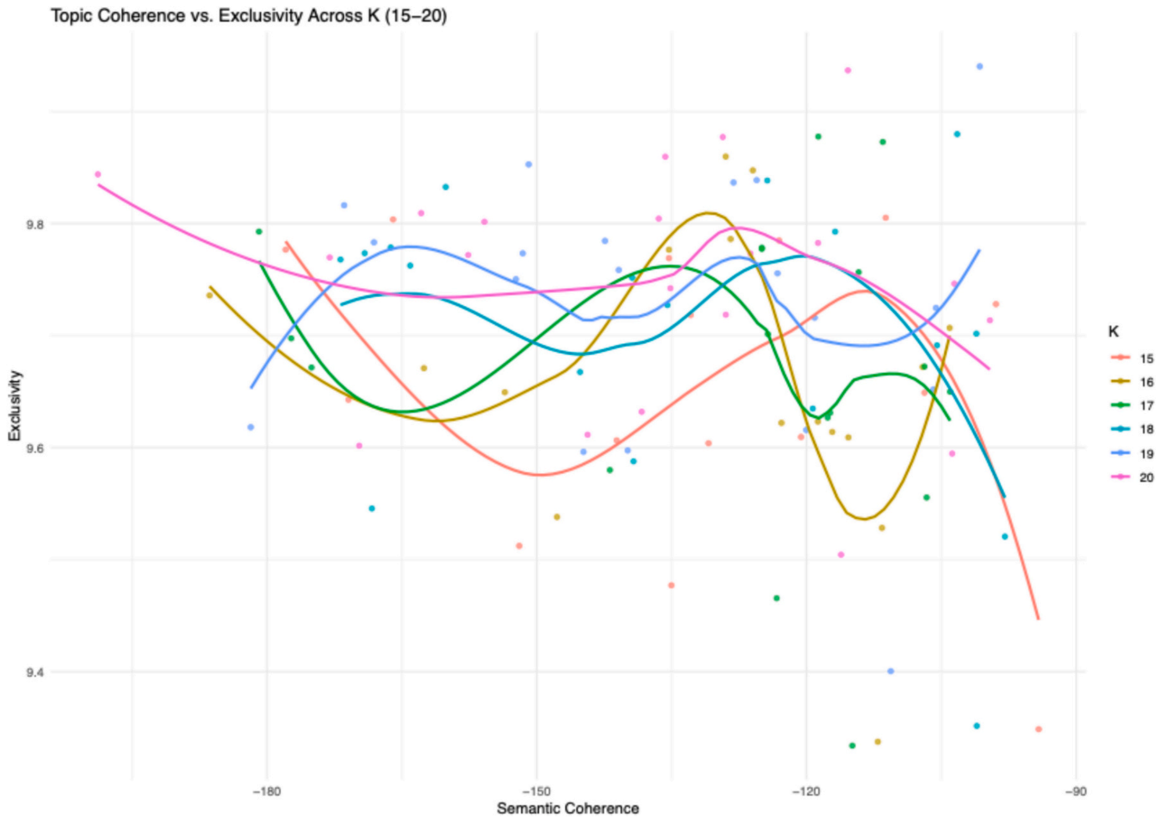

Fig. 3. Topic performance comparison.

displacement and unemployment. McKinsey Global Institute estimated that up to 375 million workers worldwide may need to switch occupational categories and learn new skills by 2030 due to automation (Manyika et al., 2017). Engineers are among the workers who may be affected by automation, as AI could automate many of the tasks they currently perform. For instance, AI could be used to design new products, conduct simulations, and optimize production processes, reducing the need for human intervention (World Economic Forum, 2018).

Topic 5 captures public discussion of ChatGPT as an emerging alternative to traditional search engines. The high-probability words "gpt," "google," "chat," "search," "bot," "internet," and "microsoft" suggest that users understand the tool in relation to existing online search and information platforms. This interpretation is reinforced by the FREX words "google," "engine," "search," "bing," "chat," "gpt," and "edge," which point to explicit comparisons between generative AI interfaces and conventional search-engine infrastructures. The topic, therefore, reflects a broader shift from keyword-based retrieval toward conversational information access. In this sense, the discussion aligns with the transition toward neural search, where users evaluate the trade-offs between the generative capabilities of large language models and the retrieval precision of traditional search engines (Zhou & Sheu, 2025).

Topic 6 presents AI as a practical instrument for increasing work efficiency, with high-probability terms such as "work" and "tool," and the FREX term "faster" pointing toward users' interest in quicker task completion. FREX terms, including "easier," "lazy," and "boring," reveal a more complex narrative. While many users appreciate the relief from repetitive or tedious activities, others express concern that frequent reliance on AI may weaken personal discipline or reduce meaningful engagement with the task itself.

Topic 7 addresses growing concerns about synthetic misinformation and the difficulty of distinguishing genuine content from AI-generated material. High-probability terms such as "misinformation" and "authenticity" signal a broad unease about the credibility of online information, while FREX terms like "deepfake," "conspiracy," and "propaganda" bring these concerns into sharper focus by pointing to concrete modes through which false or misleading material can circulate. Users often describe a sense of uncertainty about what can be trusted, suggesting that the spread of AI-produced content may weaken established cues for evaluating reliability.

Topic 8 examines questions surrounding the governance and societal consequences of releasing large-scale AI systems. High-probability words such as "system" and "public," together with FREX terms including "deployment," "impact," and "safety," indicate a conversation that moves beyond personal use cases toward the obligations of developers and institutions. Users raise concerns about how these models should be introduced, whether access should be open or limited, and what safeguards are required when probabilistic systems interact with the broader public. The topic aligns with ongoing discussions in Responsible AI and AI governance (Papagiannidis et al., 2025), where the central issues involve accountability, risk management, and the practical difficulties of ensuring reliable behavior in real-world settings.

Topic 9 reveals that users discuss "bias" through both ideological and demographic frames. The prevalence of terms, such as "political," "opinion," and "argument," in conjunction with FREX terms such as "gender," "female," "male," "conservative," and "ideology," suggests that users perceive bias as involving both political slant and unequal treatment across social identities. The especially frequent appearance of "political" suggests that ideological partiality is the more salient concern in this topic, although demographic fairness is also clearly present. This pattern suggests that public concern about bias extends beyond technical fairness diagnostics to include broader anxieties about viewpoint partiality, identity representation, and the social consequences of AI-generated judgments. It also aligns with what Eslami et al. (2019) refer to as algorithmic folk theories, where people explain AI behavior through perceived censorship or partisanship rather than through

Table 1
Topic modeling result.

Topic no.	Proportion	Top words	Topic label	Relevance
1	5.2%	Highest Prob: game, move, rule, chess, piece, play, player, strategy FREX: chess, pawn, rook, queen, stockfish, move, rule, tactic	Game Logic	Yes
2	5.9%	Highest Prob: code, work, can, get, program, error, try, script FREX: folder, file, code, fix, email, overflow, python, server,	Coding Utility	Yes
3	4.7%	Highest Prob: job, year, work, take, replace, software, tech, programmer FREX: job, engineer, industry, worker, designer, industrial, automation, software	Job Displacement	Yes
4	8.8%	Highest Prob: video, great, thank, really, love, content, amazing, see FREX: channel, joma, thomas, clearing, informative, subscribed, sponsor, video	YouTube Channel Engagement	No
5	4.9%	Highest Prob: gpt, google, chat, search, bot, internet, microsoft, bing FREX: google, engine, search, bing, chat, gpt, edge, chrome	Information Retrieval Comparison	Yes
6	10.9%	Highest Prob: make, time, need, people, get, work, think, tool FREX: easier, lazy, faster, tool, software, boring, engineer, coder	Professional Efficiency	Yes
7	3.3%	Highest Prob: social, media, misinformation, authenticity, verification, source, online, post FREX: russia, authenticity, conspiracy, deepfake, gmail, medium, bot, propaganda	Content Authenticity	Yes
8	3.7%	Highest Prob: system, important, openai, potential, issue, technology, public, model FREX: public, deployment, overall, impact, openai, potential, safety, system	Model Deployment	Yes
9	4.2%	Highest Prob: bias, opinion, political, fact, argument, people, woman, issue FREX: gender, female, political, male, conservative, ideology, identity, politic	Model Bias	Yes
10	4.9%	Highest Prob: thanks, something, comment, free, interesting, information, message, send FREX: hello, loan, message, shortlisted, testified, comment, interesting, watsapp	Spam Messages	No
11	7.3%	Highest Prob: best, trading, market, expert, profit, trade, investment, free FREX: forex, trading, crypto, bitcoin, broker, trade, investing, profit	Investment Scams	No
12	2.8%	Highest Prob: writing, school, teacher, essay, written, teach, learning FREX: student, grade, assignment, teacher, homework, mobile, essay	AI-Enhanced Education	Yes
13	6.3%	Highest Prob: world, life, change, people, live, future, society, humanity FREX: humanity, concern, threat, advancement, inevitable, artificial, control, existence	Safety Risks	Yes
14	2.8%	Highest Prob: prompt, response, reply, ask, generate, say, instruction, output, FREX: jailbreak, jailbroken, exploit, bypass, override, unrestricted, persona, forbidden	AI Jailbreaking	Yes
15	6.1%	Highest Prob: science, math, answer, question, wrong, calculation, information, mistake FREX: physics, equation, miscalculated, symbolic, computation, reasoning, incorrect, answer	Inaccuracy in STEM	Yes
16	5.8%	Highest Prob: data, model, language, understand, human, learning, knowledge, trained FREX: neural, trained, eigenvalue, train, dataset, algorithm, network, logic	AI Architecture	Yes
17	3.2%	Highest Prob: year, old, ago, month, decade, tech, change, evolution FREX: doctor, genai, year, linus, diffusion, past, developing, different	Tech Evolution	Yes
18	2.8%	Highest Prob: day, always, everything, every, never, behind, started, left FREX: behind, jealous, achievement, became, day, congratulate, everything, approval	Personal Anecdotes	No
19	3.7%	Highest Prob: write, generate, result, text, create, review, help, accurate FREX: plagiarism, detector, originality, article, generic, formulaic, unreliable, unoriginal	Creative Writing	Yes
20	2.9%	Highest Prob: good, bad, people, perfect, wrong,	Ethical Judgment	Yes

(continued on next page)

Table 1 (*continued*)

Topic no.	Proportion	Top words	Topic label	Relevance
		harm, fair, nothing FREX: bad, evil, choice, perfect, moral, ethical, dangerous, stupid		

underlying data processes. While some comments raise broader fairness concerns, the dominant anxiety is that AI may shape public discourse by favoring certain viewpoints over others. Unlike the conventional technical understanding of algorithmic bias, which emphasizes distortions linked to gender, racial, or economic imbalances in data (Ntoutsi et al., 2020), the pattern observed here highlights the importance of perceived ideological bias in public evaluations of generative AI. Survey evidence likewise suggests that many individuals view political bias as a major risk of AI-generated content (Peters, 2022).

Topic 12 captures public discussion of ChatGPT as a tool in educational settings. The high-probability words “school,” “teacher,” “essay,” and “learning” suggest that the topic is situated within formal education rather than general everyday use. This interpretation is further supported by the FREX words “student,” “grade,” “assignment,” “homework,” and “essay,” which highlight concrete academic activities and classroom responsibilities. Overall, the topic reflects how users perceive ChatGPT's growing role in education, especially in supporting assignments, assessment-related tasks, and teacher–student academic interactions.

Topic 13 reflects macro-level existential anxieties. This topic discusses the broader concerns about the long-term consequences of AI development. The top words link “future” and “humanity” (Highest Prob) with risk-oriented FREX terms like “inevitable,” “threat,” and “existence.” This narrative mirrors the debate about whether advanced Artificial General Intelligence systems can be reliably aligned with human interests or if their advancement poses a fundamental, existential risk to human agency (Cheng & Gong, 2024).

Topic 14 focuses on adversarial user behaviors, which describe attempts to bypass safety filters and elicit restricted outputs. High-probability terms “prompt” and “response” are contextualized by FREX terms “jailbreak,” “bypass,” and “override.” These comments illustrate a recognizable pattern in security and human–AI interaction research, where users actively probe for weaknesses in rule-based or probabilistic control layers. Such behaviors raise important questions about the durability of safeguard designs in open-ended language models, as well as the extent to which conversational systems can maintain consistent normative boundaries when facing adversarial prompting.

Topic 15 emphasizes the existing gap between the model's ability to generate linguistically fluent content and its capacity to ensure factual accuracy. High-probability terms “science” and “math” are reinforced by FREX terms “physics,” “equation,” and “miscalculated,” findings that while models excel at syntax, they often fail at symbolic computation and logical reasoning, leading to plausible but factually incorrect outputs.

Topic 16 reflects sustained user interest in the internal mechanics of large language models. The high-probability words “data,” “model,” “language,” “learning,” and “knowledge” suggest that users are concerned with how these systems process information and generate responses. This interpretation is reinforced by the FREX words “trained,” “dataset,” “algorithm,” “network,” and “logic,” which point more directly to the technical foundations of model training and operation. Overall, the topic captures attempts to understand how large language models learn from data and produce language-based outputs.

Topic 17 situates GenAI within broader narratives of rapid technological advancement. High-probability words such as “month,” “year,” and “change,” alongside FREX terms like “genai,” and “diffusion,” frame the discussion in terms of technological change over time. Many comments compare current AI technology with its predecessors, highlighting both the transformative nature of this new wave and its perceived superiority over earlier iterations. This reflective discourse often casts GenAI as part of a larger pattern of accelerating technological progress, one that is distinct not just in terms of capability but also in its speed and societal impact.

Topic 19 reflects user evaluations of the model's ability to produce original creative content, with many critiques focused on the perceived derivative or repetitive nature of its outputs. High-probability terms such as “generate” and “create” signal general content-generation tasks, while FREX terms including “plagiarism,” “formulaic,” and “unoriginal” point to concerns about authenticity and creative value. These discussions connect closely to debates in computational creativity, particularly the view that generative systems often assemble material by drawing on familiar patterns present in their training data rather than producing fully original work (De Cremer et al., 2023). The topic also intersects with concerns in copyright scholarship, as users question the legitimacy of creative outputs built from large, scraped text corpora.

Topic 20 reflects users' tendency to assess AI systems using explicit moral frameworks, often categorizing the system's behavior in binary terms such as “good,” “bad,” or “harm.” The frequent appearance of high-probability terms like “good” and “harm,” alongside FREX terms such as “evil,” “moral,” and “ethical,” suggests that users are not only evaluating the system's functional performance but are also engaging in higher-order ethical reasoning. This topic is closely tied to studies in machine ethics and anthropomorphism, where users attribute moral agency to AI systems (Sullivan & Fosso Wamba, 2022). In this context, users probe whether the system's actions align with human standards of benevolence, responsibility, and ethical behavior, questioning if AI can be trusted to act in ways that reflect human values and norms.

4.3. Topic distribution analysis

The analysis of topic prevalence reveals a structural contrast in user experience, where opinions differ markedly depending on the type of interaction. Fig. 4 shows that positive sentiment is overwhelmingly driven by instrumental utility, with topics such as ‘Professional Efficiency,’ ‘Coding Utility,’ and ‘AI-Enhanced Education’ appearing frequently in favorable reviews. This pattern suggests that users value the model most when it functions as a productivity multiplier and as a catalyst for technological advancement. In contrast, negative sentiment concentrates on epistemic failures and societal anxieties, particularly in ‘Job Displacement,’ ‘Inaccuracy in STEM,’ and ‘Safety Risks.’ This distribution indicates that dissatisfaction arises not from general hostility but from specific experiences involving unreliability, perceived bias, or economic threat. Positioned between these poles are technical themes such as ‘AI Architecture’ and ‘Model Deployment,’ which cluster near zero, reflecting a neutral or exploratory discourse that lacks strong evaluative judgment. Overall, these findings demonstrate that user sentiment reflects reactions to concrete experiential domains: users celebrate the system as a practical aid while simultaneously critiquing it as a source of error and wider disruption.

4.4. Topic correlation analysis

To examine how discussion themes co-occurred within the corpus, we estimated topic correlations using the STM package's *topicCorr* function in R. This procedure derives a topic correlation structure from the model-estimated topic proportions across documents and identifies conditional associations among topics that tend to appear together within the same comments. To enhance interpretability and eliminate visually weak associations, only topic pairs with correlation coefficients exceeding 0.4 were included in the final visualization, whereas node size is scaled according to topic proportion. This thresholding approach allows the network to highlight the stronger structural relationships

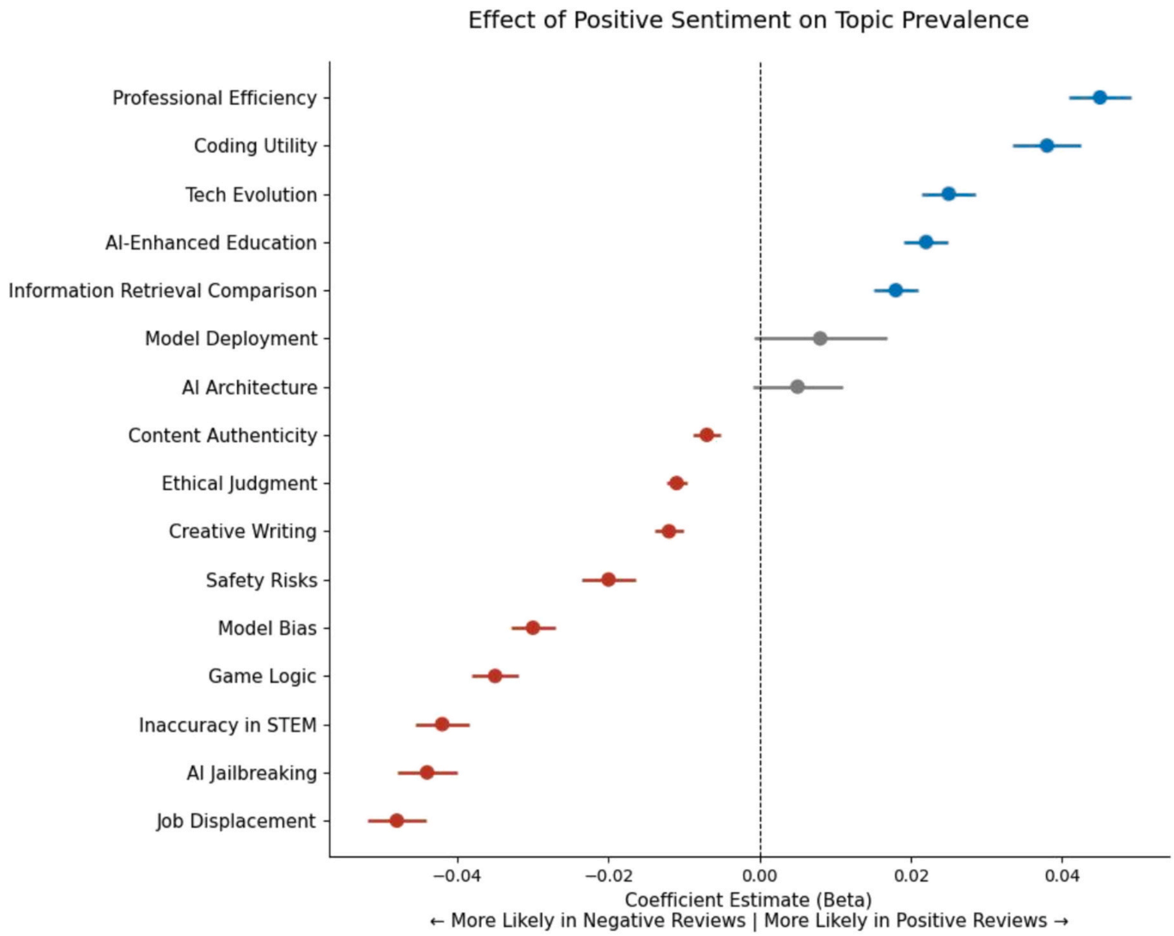

Fig. 4. Topic distribution across positive and negative sentiments.

among themes while reducing noise from marginal links.

Fig. 5 reveals that user discourse is not driven by a single dominant narrative but rather organizes into three coherent thematic clusters. Examining the proximity of topics within the network reveals the reasoning that connects issues that might otherwise appear unrelated. At the center is a cluster linking 'AI Architecture,' 'Model Deployment,' 'AI Jailbreaking,' and 'Inaccuracy in STEM.' Their close association suggests the presence of a technically oriented user group that investigates the system's internal workings, encounters its functional constraints, and experiments with ways to push or circumvent those boundaries. This pattern illustrates how technical curiosity and adversarial testing often coexist within the same community. Surrounding this core is a second cluster reflecting concerns about the social and economic implications of widespread AI adoption. Topics such as 'Professional Efficiency' and 'Job Displacement' frequently co-occur, capturing the tension between using the system to enhance productivity and recognizing its potential to disrupt employment. The presence of 'Safety Risks' and 'Ethical Judgment' within this grouping shows that users evaluate practical benefits alongside concerns about harm, responsibility, and the authenticity of AI-generated content. A third cluster brings together 'Creative Writing' and 'AI-Enhanced Education.' Unlike the technical and societal clusters, this grouping focuses on collaborative or instructional uses in which AI acts as a support mechanism. Users in this cluster describe the system less as an autonomous actor and more as a partner that extends creative or learning capacities. Its relative separation suggests that individuals engaged in creative or educational tasks experience AI as a tool for augmentation rather than substitution or structural disruption.

5. Discussion and conclusion

5.1. Discussion of major findings

The findings of this study indicate that the general public's assessments primarily center on the reliability and performance of ChatGPT, accompanied by concerns about its societal impacts and AI bias. Within the context of rapid innovation in intelligent content creation, ChatGPT stands out as a key contributor to refined user engagement and technological performance, though its deployment raises various implementation difficulties and ethical concerns.

The research revealed several consistent patterns concerning ChatGPT's capabilities, such as users' challenges, doubts about dependability, comparisons with other AI engines, performance appraisals, and investigations into its fundamental principles. These findings are consistent with recent work by Wu et al. (2023) and Roumeliotis and Tselikas (2023), which indicate a growing interest in assessing ChatGPT's capabilities. In addition, our analysis confirms Wang et al.'s (2023) findings on its skill in language-based operations, including translation, text production, and conversational design. These attributes suggest that ChatGPT can serve as an effective means of elevating productivity and correctness in numerous workplace contexts. However, our findings also reveal persistent user concerns about the accuracy and reliability of ChatGPT when handling complex queries, particularly in comparison to AI products from tech giants like Google and Microsoft. This reliability issue is consistent with research by Floridi (2023) and Yang and Zhang (2024), which suggests that GenAI systems may produce inaccurate responses when they fail to fully grasp user intent, a limitation directly linked to the quality and scope of their training data.

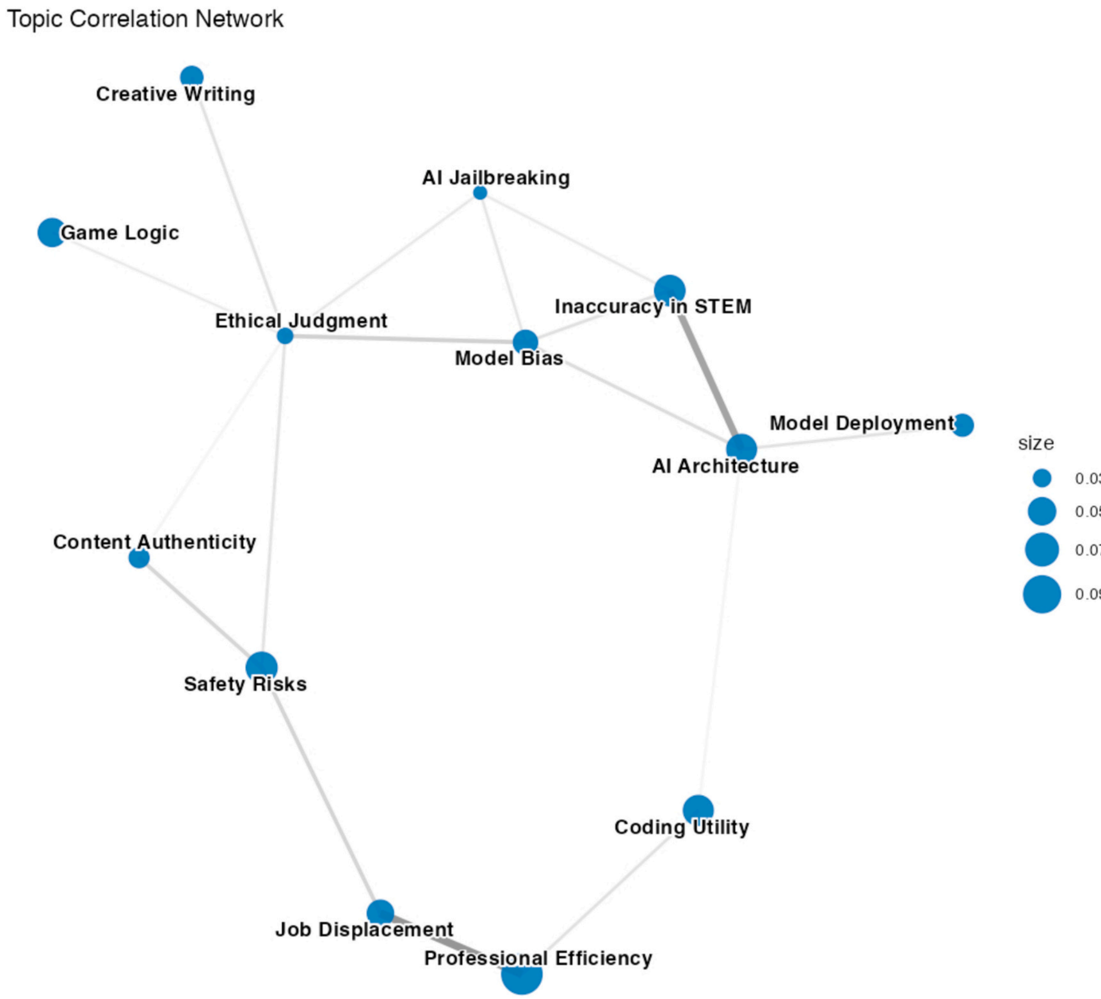

Fig. 5. Topic correlation map.

Our findings further support the observations of Nazir and Wang (2023) and Keshamoni (2023), indicating that although ChatGPT provides a more engaging interaction experience than many competing AI systems, this advantage is moderated by concerns about data bias and inconsistent response quality. Yet these limitations do not prevent users from comparing it favorably to traditional search engines. Many view ChatGPT as offering a more flexible and personalized way of accessing information, even when inaccuracies occur. As one commenter remarked, "*Chat gpt is like a very very personalised search engine, if they launch it as a search engine, Google chrome will die,*" reflecting a broader perception that ChatGPT represents a fundamentally different mode of information retrieval.

The findings of this study reveal that public concerns about ChatGPT primarily center on three areas: societal impacts, job market effects, and AI biases. From a societal impact perspective, our findings suggest that while ChatGPT and similar AI technologies offer significant benefits in terms of convenience and enhanced interaction efficiency, they also pose potential risks. Labadze et al. (2023) suggest that over-reliance on these technologies could lead to an erosion of social skills and empathy, a view echoed in our qualitative analysis, where users warned that "*don't let something on a screen change who you are*" could lead to a loss of self-identity. Building on this, our findings align with Yekta's (2024) assertion that the widespread adoption of AI chatbots could contribute to increased social isolation. A significant issue is the possibility that individuals will opt for machine-mediated interaction over direct human contact, which may harm mental health and alter established social dynamics. Regarding the job market, our findings align with the broader debate on AI-driven labor disruption. Consistent with Frey and Osborne (2017), users continue to worry that AI may displace low-skill and repetitive forms of work. However, the comments suggest that these concerns have expanded beyond traditionally vulnerable sectors. Engineers, designers, programmers, and other knowledge workers increasingly express uncertainty about the durability of their roles as GenAI capabilities advance. As another user noted, "*A.I will take away all the creative jobs... It's won't erase it, but only the top of the field will survive,*" highlighting a growing belief that even high-skill professions are no longer insulated from technological change.

At the same time, this study reveals that AI bias is a significant ethical concern. Existing research has shown that AI bias can emerge from incomplete or skewed training data and may reproduce socially embedded inequalities (Yang & Zhang, 2024). The pattern identified in the dataset of this research indicates a greater concern regarding political or ideological bias than mere demographic inequity. The prominence of the term "political" in both the Highest Prob and FREX terms for Topic 9 suggests that users frequently interpret biased outputs as signs of ideological slant or selective framing. This finding is consistent with Peters (2022), who argues that algorithmic bias may also operate through political orientation and can be particularly difficult to detect and correct because social norms against political bias are typically weaker than those against gender or racial bias. Accordingly, our findings suggest that public evaluations of ChatGPT are shaped not only by concerns about technical fairness but also by fears that AI systems may embed or amplify ideological preferences in ways that influence trust and public discourse.

5.2. Implications

The findings of this study contribute to several areas of research by offering a clearer understanding of how the public perceives ChatGPT and how these perceptions relate to debates on AI adoption. Through the use of topic modeling on large-scale UGC, the study offers an analysis of the strengths and limitations that users associate with ChatGPT, including context-specific accuracy, authenticity of generated content, and alignment with ethical expectations. Collectively, these observations expand ongoing work in HCI by showing not only how individuals engage with AI systems but also how their evaluations cluster into distinct themes that reflect different forms of interaction and expectation. The combination of topic prevalence, sentiment structure, and topic correlations demonstrates that public evaluations are organized along coherent lines of reasoning, which allows for more grounded theoretical development within AI and society research. The methodological contribution lies in demonstrating that topic modeling can reveal both topic content and the structural relationships among topics, enabling a systematic analysis of discourse patterns in unstructured, real-time data (Roberts et al., 2016).

These findings also carry practical implications. Developers can use the domain-specific insights to guide improvements in key areas such as STEM factual accuracy, code validation, and mechanisms that enhance content authenticity. The technical cluster identified in the topic network indicates that users who explore system capabilities in greater depth are also those who encounter and report the system's limitations. This shows the importance of making protections against adversarial manipulation stronger and of making models more transparent so that users who want clearer explanations of how the system works can obtain them. The creative-educational cluster suggests that users value ChatGPT as a collaborative tool in learning and creative development. This illustrates the importance of culturally sensitive design choices, including educational features that support learning while avoiding over-dependence and creative tools that balance assistance with originality. Policymakers can draw on these findings to shape governance frameworks that address public concerns about ethics, privacy, and fairness. The analysis of societal effects, such as challenges to content integrity, safety considerations for the future, and changes in workforce requirements, supplies a data-supported basis for developing appropriate governance measures.

5.3. Limitations and suggestions for future research

This study has several limitations. First, YouTube served as the primary platform for data collection, and although it hosts a wide range of users, its breadth introduces variability that may obscure domain-specific insights. Future research could benefit from data gathered in specialized forums, coding communities, and educational platforms for a more focused examination of user perceptions. Second, this study does not fully account for cultural context. Cultural norms influence how users assess AI's ethical implications, biases, and impact on sectors like education and employment (Zhang et al., 2025). Cross-cultural research is needed to explore how perceptions of ChatGPT vary globally and how these insights can guide AI development for diverse populations. Additionally, this study relied on textual content from YouTube comments, which may not reflect the entire range of user interactions with ChatGPT. Future research should incorporate multimodal data, such as video-based feedback or user interactions on other platforms, to capture a broader range of user experiences. Lastly, UGC may reflect self-selected, biased opinions. To achieve a more representative understanding of public perception, future studies should consider combining YouTube data with surveys or focus groups to capture a broader demographic.

CRediT authorship contribution statement

Kai Ding: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Yan Yang:** Writing – review & editing, Supervision, Investigation, Formal analysis. **Dongwei Zhao:** Writing – original draft, Resources, Methodology, Data curation. **Xiaoqian Sun:** Visualization, Validation, Methodology. **Lin Wang:** Investigation, Data curation.

Ethical considerations

Ethical review and approval were not applicable because the data are publicly available and non-identifiable. This study does not involve human participants. We analyzed user-generated comments that are publicly available on YouTube. No interaction or intervention occurred, and no identifiable personal information was collected or retained.

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Declaration of competing interest

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Data availability

Data will be made available on request.

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